LLaDA2.0-Uni: Unifying Multimodal Understanding and Generation with Diffusion Large Language Model

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Research Questions

- How can we create a unified architecture that performs both multimodal understanding and generation using discrete diffusion language models instead of autoregressive approaches?
- Can semantic tokenizers (SigLIP-VQ) overcome the limitations of reconstructive VQ tokenizers in unified multimodal models while maintaining generation quality?
- How can we enable interleaved generation and reasoning capabilities within a single unified diffusion-based framework?

Core Idea

Current unified multimodal models suffer from a fundamental mismatch: they use reconstructive VQ tokenizers that compress images for generation but lose semantic information needed for understanding. LLaDA2.0-Uni's key insight is to use SigLIP-VQ, a purely semantic tokenizer trained on understanding tasks, which preserves rich semantic information while still enabling discrete diffusion modeling. The architecture combines this semantic tokenizer with a 16B MoE diffusion LLM backbone that processes both text and visual tokens under unified block-level masked diffusion, plus a separate diffusion decoder that reconstructs semantic tokens into high-fidelity images. This design eliminates the modeling gap between understanding and generation while enabling parallel decoding and interleaved reasoning capabilities.

Key Formula

SPRINT Importance Scoring

$$s_i = \alpha \cdot \bar{I}_i + (1-\alpha) \cdot c_i$$

Combines key-norm importance \bar{I}_i (how strongly position i influences attention) with token confidence c_i (model's prediction certainty) to score prefix positions for pruning. α=0.5 balances the two measures to retain most informative tokens during inference acceleration.

Mask Token Reweighting Loss

$$\mathcal{L}_{MTRS} = \frac{\sum_j \beta_j \mathcal{L}_{SFT}^{(j)}}{\sum_j \beta_j}, \text{ where } \beta_j = \frac{1}{\sqrt{\sum_k \sum_i 1[x_{t,k,i}^{(j)} = [MASK]]}}$$

Reweights SFT loss by inverse square root of masked tokens per sample j to balance gradient contributions across samples with vastly different lengths (up to two orders of magnitude). Prevents long sequences from dominating gradients while avoiding brevity bias.

Key Figure

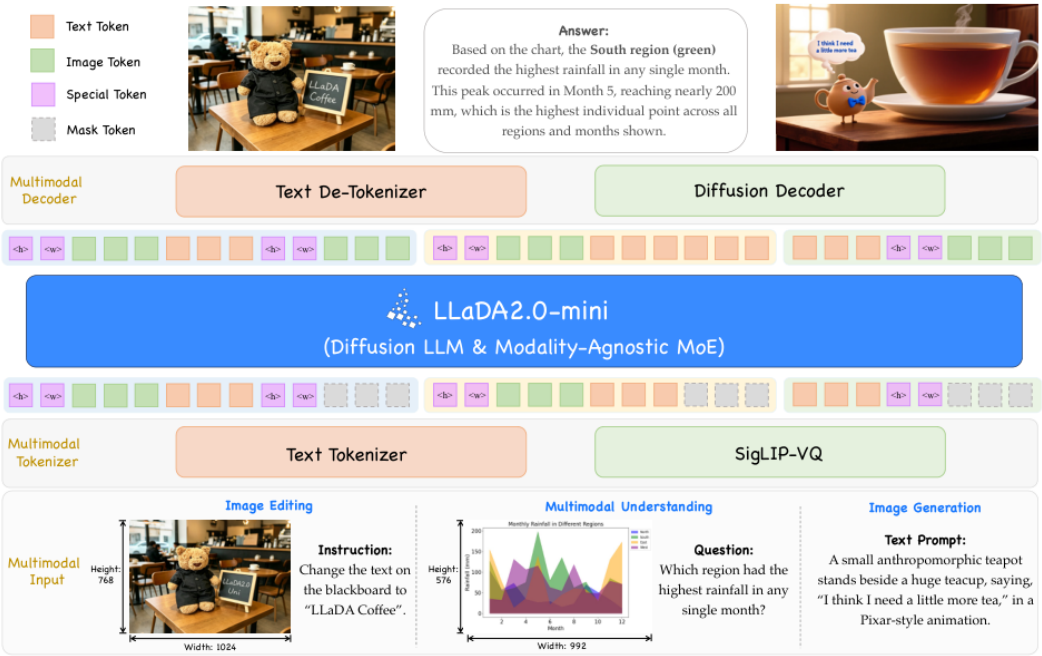

Figure 4: Architecture Overview of LLaDA2.0-Uni. The framework integrates a SigLIP-VQ tokenizer and a language model decoder to process multimodal inputs, including text, image, and video.

Figure 4: Shows the complete unified architecture with SigLIP-VQ tokenizer, dLLM backbone with modality-agnostic MoE, and diffusion decoder processing multimodal understanding, generation, and editing tasks through shared discrete token representation.

Limitations

- SigLIP-VQ tokenizer struggles to preserve fine-grained image details needed for detail-sensitive tasks like image editing
- Interleaved generation and reasoning capabilities require further scaling of training data and model capacity to reach full potential
- Text rendering performance falls short of leading specialized models, indicating area for improvement
- Reinforcement learning optimization for unified dLLMs remains challenging and requires further refinement